



SECOND SEMESTER 2025-26
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : ME G511

Course Title : Mechanism and Robotics

Instructor-in-Charge : Dr. Prateek Kala

1. Scope and Objective of the Course:

This course is designed to provide an in-depth and holistic understanding of the technology and principles underlying modern robotics. Over the years, the demand for human-like machines capable of replacing or assisting humans in diverse and challenging work environments has grown significantly due to reasons such as efficiency, precision, safety, and the need to operate in hazardous conditions. In response to this need, the field of robotics has rapidly evolved and established itself as one of the most critical and promising areas of engineering for the future.

The course aims to build a strong interdisciplinary foundation for students, focusing particularly on the mechanical aspects of robotic systems while integrating relevant knowledge from electronics, computer science, and control engineering. It will cover fundamental and advanced concepts related to mechanisms used in robotic systems, their unique characteristics, and detailed kinematic and dynamic analysis essential for design and development. Furthermore, students will gain insights into the design methodologies, practical implementation principles, and real-world industrial applications of robotics. The course will also explore current issues, trends, and innovations in robotics, enabling students to appreciate its impact across industries and prepare them for future technological advancements in this rapidly expanding domain.

Text Book:

(T1) Mittal R. K. & Nagrath, I. J., “Robotics and Control”, TMH, 2003 (Reprint 2010 or later).

Reference Books:

(R1) Groover, M. P., et al., “Industrial Robotics”, McGraw Hill ISE, 1986.

(R2) Fu, K. S., et al., Robotic: Control, Sensing, Vision & Intelligence, McGraw Hill ISE, 1987

(R3) Robert J., Schilling, Fundamentals of Robotics: Analysis and Control, Prentice Hall, NJ, 2002.





4. Course Plan

Module	Topics Covered (Detailed)	No. of Lectures
Module 1: Introduction to Mechanisms and Robotics	History and evolution of robotics; Difference between mechanisms, machines, and robots; Classification of robots (industrial, service, humanoid, mobile); Applications in manufacturing, healthcare, defense, and space	4
Module 2: Fundamentals of Mechanisms	Basic terminology (links, joints, kinematic pairs); Degrees of freedom (DOF) and mobility; Gruebler's equation; Planar vs. spatial mechanisms; Four-bar linkages and coupler curves	4
Module 3: Kinematics of Robotics	Forward kinematics and transformation matrices; Denavit-Hartenberg (D-H) parameters (standard & modified); Frame assignment rules; Inverse kinematics (analytical & numerical methods); Workspace analysis and redundancy	8
Module 4: Differential Kinematics & Jacobians	Differential motion and velocity propagation; Jacobian matrix derivation for serial robots; Manipulability and velocity ellipsoids; Singularities and their impact on control	4
Module 5: Dynamics of Robots	Newton-Euler and Lagrangian formulations; Forward and inverse dynamics problems; Dynamic modeling of manipulators; Payload capacity, torque requirements, and force analysis	4
Module 6: Robot Control	Fundamentals of robot	4





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	control systems; Joint space (PID) control strategies; Task space control (computed torque, adaptive control); Hybrid position-force control	
Module 7: Advanced Topics in Robotics	Mobile robot kinematics and navigation; Path and trajectory planning (C-space, PRM, RRT); Robot vision (camera calibration, image processing); AI and machine learning integration in robotics	6
Module 8: Industrial Robotics & Applications	Industrial robot configurations (SCARA, Cartesian, articulated); Robot programming methods (teach pendant & offline); End-effectors and grippers design; Case studies of robotic automation in industry	4
Module 9: Emerging Trends & Project Work	Collaborative robots (cobots) and human-robot interaction; Soft robotics and bio-inspired robots; Swarm robotics; Course review and student project presentations	2

5. Evaluation Scheme:

EC No.	Evaluation Component	Duration	Weightage	Date, Time & Venue	Nature of Component
1	Mid Sem Test	90 min	30	Refer Time table	CB*
2	Surprise Tests/ Quizzes / Assignments/ Projects and seminars	-	35	Deadlines will be informed in class	
3	Comprehensive Examination	3Hrs	35	Refer Time table	CB/OB

6. Chamber consultation hour: To be announced in the class.

7. Notices: All notices regarding the course will be displayed only on Mechanical Engineering Department



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notice board.

8. Make-up Policy: Make-up will be granted ONLY in genuine cases with prior permission. The request application for make-up test MUST be reached to the Instructor-in-Charge before commencement of the scheduled test along with DOCUMENTARY PROOF. No make- up will be allowed for the Surprise Quiz Tests.

Instructor-in-Charge,

ME G511



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